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#include <stdio.h>

#define MAX_DEGREE 10

void read_polynomial(int degree, int poly[]) {
    for (int i = 0; i <= degree; i++) {
        printf("Enter the coefficient for x^%d: ", i);
        scanf("%d", &poly[i]);
    }
}

void print_polynomial(int degree, int poly[]) {
    for (int i = degree; i >= 0; i--) {
        if (poly[i] != 0) {
            printf("%d", poly[i]);
            if (i > 0) {
                printf("x^%d", i);
            }
            if (i > 0 && poly[i - 1] >= 0) {
                printf("+");
            }
        }
    }
    printf("\n");
}

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void add_polynomials(int degree, int poly1[], int poly2[], int result[]) {
    for (int i = 0; i <= degree; i++) {
        result[i] = poly1[i] + poly2[i];
    }
}

int main() {
    int d1, d2;

    printf("Enter the degree of the first polynomial: ");
    scanf("%d", &d1);

    printf("Enter the degree of the second polynomial: ");
    scanf("%d", &d2);

    if (d1 < 1 || d1 > MAX_DEGREE || d2 < 1 || d2 > MAX_DEGREE) {
        printf("Invalid input. Degree must be between 1 and %d.\n",
MAX_DEGREE);
        return 1;
    }

    int poly1[d1 + 1], poly2[d2 + 1], result[(d1 > d2 ? d1 : d2) + 1];

    printf("Enter the first polynomial:\n");
    read_polynomial(d1, poly1);

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printf("Enter the second polynomial:\n");
read_polynomial(d2, poly2);

int max_degree = (d1 > d2) ? d1 : d2;
add_polynomials(max_degree, poly1, poly2, result);

printf("First polynomial: ");
print_polynomial(d1, poly1);

printf("Second polynomial: ");
print_polynomial(d2, poly2);

printf("Resultant polynomial: ");
print_polynomial(max_degree, result);

return 0;
}

```

## OUTPUT

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Enter the degree of the polynomial: 4
Enter the first polynomial:
Enter the coefficient for x^0: 4
Enter the coefficient for x^1: 2
Enter the coefficient for x^2: 3
Enter the coefficient for x^3: 0
Enter the coefficient for x^4: 7
Enter the second polynomial:
Enter the coefficient for x^0: 0
Enter the coefficient for x^1: 0
Enter the coefficient for x^2: 4
Enter the coefficient for x^3: 5
Enter the coefficient for x^4: 2
First polynomial: 7x^4+3x^2+2x^1+4
Second polynomial: 2x^4+5x^3+4x^2+
Resultant polynomial: 9x^4+5x^3+7x^2+2x^1+4

```